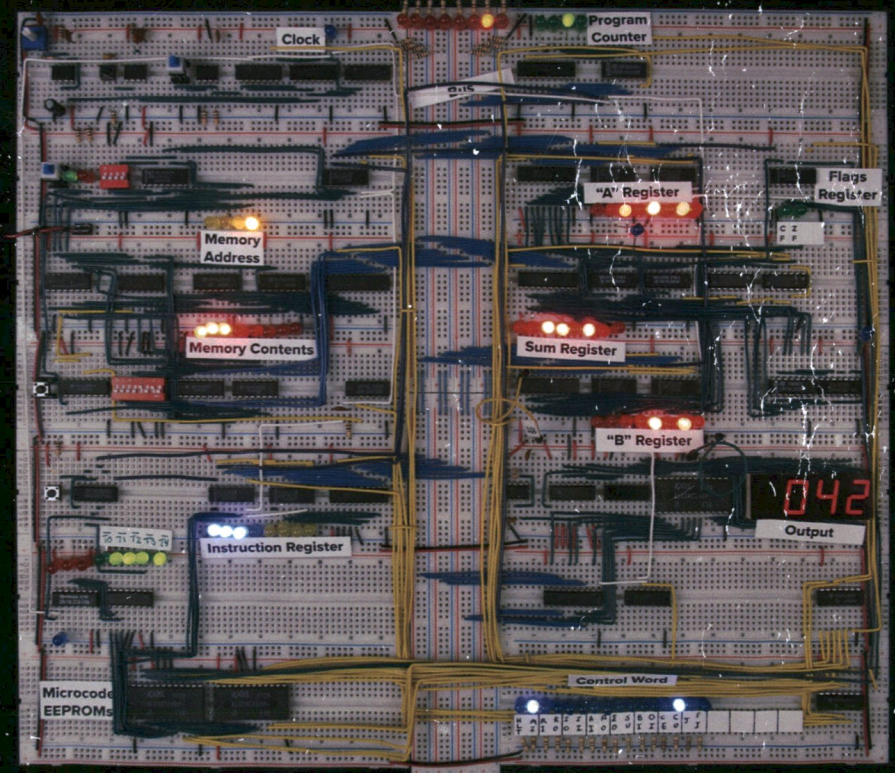
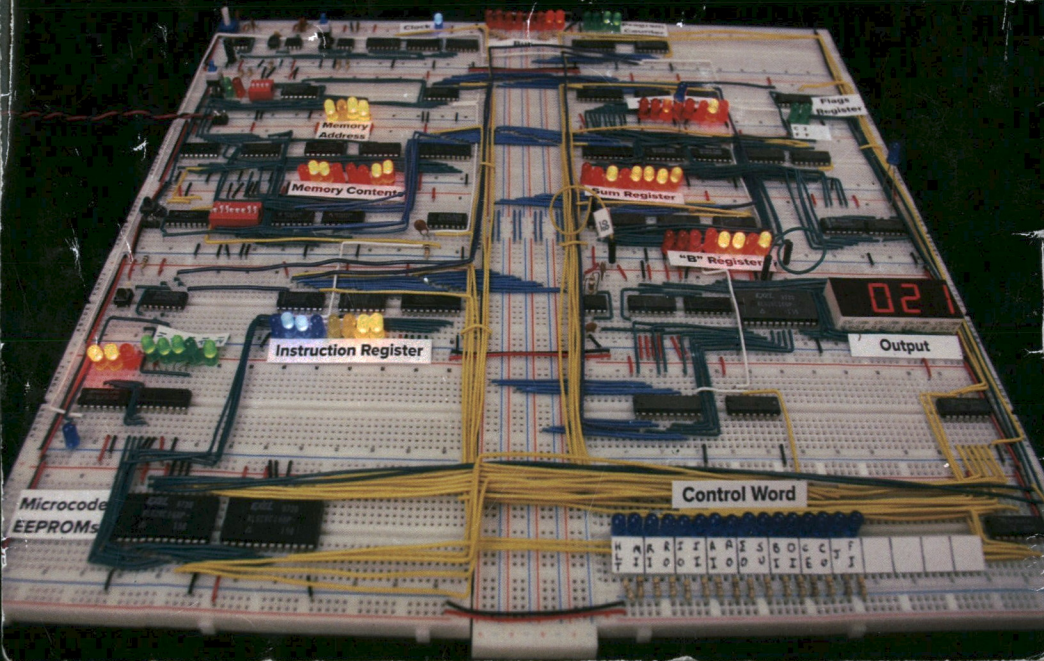


8-bit breadboard computer

Kit 4 of 4

Output and control logic

www.eaten.net
youtube.com/beneater



Ben Eater

Welcome!

This is the final kit in a set of four that, along with my series of videos on YouTube, will help you develop a deep understanding of how programmable CPUs really work. I hope when you're done, you'll know enough to extend the computer design and make it your own.

With this final kit, you'll build the output display and control logic and assemble everything from all four kits into the full programmable computer! The modularity of the design makes it easier to test each module by itself so you shouldn't ever get to a point where you put it all together and nothing works.

The output register is similar to any other register (like the A and B registers) except rather than displaying its contents in binary on 8 LEDs, it displays its contents in decimal on a 7-segment display.

Finally, the control logic is the heart of the CPU. It's what defines the opcodes the processor recognizes and what happens when it executes each instruction.

Where are the instructions to build this kit?

Surprise! This booklet doesn't actually contain the full instructions for building the kit. This kit is a companion to my series of YouTube videos. You should watch the videos and follow along yourself. I hope that by *not* having a detailed "paint-by-numbers" instruction book, you'll learn more about how the circuits really work:

The video instructions for this kit are at

www.eater.net/8bit/output

www.eater.net/8bit/bus

www.eater.net/8bit/control

Arduino EEPROM programmer

In the videos, we build a programmer for the 28C16 EEPROM using an Arduino. All of the parts to build the programmer are included in this kit. You can find the code used in the videos from the following GitHub repository:

<https://github.com/beneater/eeprom-programmer>

